Psicologia: Reflexão e Crítica

RESEARCH | Article in Press

Development of the facial emotion recognition task (TAREFE): methodological aspects and initial psychometric indicators

Flávia de Lima Osório · Diego Alves Ferreira · Nelson Torro-Alves

Received: 14 February 2026 / Accepted: 24 August 2026
© The Author(s) 2026

Abstract

Background Facial Emotion Recognition (FER) is a core component of social cognition, supporting theory of mind processes, empathy, and social reasoning. Deficits in FER are associated with functional impairments across multiple domains of typical and atypical development. Measurement instruments with sound psychometric properties are therefore essential to support clinical assessment in different contexts.

Objective To present the development process of the "Facial Emotion Recognition Task" (TAREFE), a computer-based test designed to assess FER in Brazilian adults, with emphasis on the methodological procedures and the initial psychometric indicators.

Methods A set of stimuli depicting Brazilian faces was used (n = 170, with diversity in sex and racial background). The stimuli represented the six basic emotions and a neutral expression and had been previously validated in a general population sample. Stimulus selection was based on Item Response Theory (1PL and 2PL), with analysis of infit and outfit indices, S-X^2 analysis, item difficulty (b) and discrimination (a).

Results Twenty-eight stimuli were selected, showing excellent discrimination (46% moderate; 54% high or very high) and varied levels of difficulty (43% easy or very easy; 39% moderate; 18% difficult). None of the selected stimuli showed a misfit to the model. In TAREFE, participants are required to identify the emotion expressed by each face by choosing one of the seven response alternatives (Stage 1). Accuracy rates and response times are automatically recorded. To enhance task discriminatory power, an additional optional stage (Stage 2) was introduced, in which stimuli are presented at lower intensity levels (50% and 70% of the target emotion, generated using morphing techniques) to increase recognition difficulty.

Keywords Emotion Recognition · Facial Expression · Social Cognition · IRT · Emotions · Psychometrics

Abbreviations

FER	Facial Emotion Recognition
ToM	Theory of Mind
ASD	Autism Spectrum Disorder
TAREFE	Tarefa de Reconhecimento de Expressão Facial
BMPAT	Bootstrap Modified Parallel Analysis Test

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.



https://doi.org/10.1186/s41155-026-00410-4

Published online: 16 October 2026

Springer

POFA	Pictures of Facial Affect
DANVA–2	Diagnostic Analysis of Nonverbal Accuracy–2nd Edition
SATEPSI	*Sistema de Avaliação de Testes Psicológicos* (Psychological Test Assessment System)
CFP	Brazilian Federal Council of Psychology
DIF	Differential Item Functioning
IRT	Item Response Theory
AL	Joy
ME	Fear
RA	Anger
SU	Surprise
TR	Sadness
NO	Disgust
NE	Neutral
BA	Open mouth
BF	Closed mouth
1PL	One-parameter Logistic Model
2PL	Two-parameter Logistic Model
ICCs	Item Characteristic Curves

Introduction

Facial emotion recognition (FER) constitutes one of the primary means of nonverbal communication in humans (Kaulard et al., 2012). According to Darwin (1872/1998), facial expressions are universal and innate, with evolutionary roots and adaptive functions essential for survival, a view that motivated his extensive investigations (Ekman, 2009).

As a core component of social cognition—the multifaceted set of processes that enable the perception, interpretation, and response to social cues—FER is the ability to correctly identify and categorize emotional states from nonverbal visual information. This ability is not a passive perceptual process but an active one that is foundational to higher-level social cognitive functions, including mentalization (Theory of Mind; ToM), empathy (both affective and cognitive), and social reasoning (Huang et al., 2025; Lee et al., 2014; Lin et al., 2022; Oliver et al., 2019; Quinting et al., 2022).

Deficits in FER skills are associated with functional impairments across multiple domains of typical and atypical development, with robust clinical implications in conditions such as autism spectrum disorder, schizophrenia, mood disorders, and neurocognitive disorders. These impairments are relevant for diagnostic assessment, therapeutic planning, and clinical monitoring (Gao et al., 2021; He et al., 2012; Krause et al., 2021; MacPherson et al., 2021; Muñoz et al., 2021; Revilla et al., 2025; Thonse et al., 2018). In this context, valid and accurate measures of FER are essential to support clinical assessments across different settings.

The pioneering studies by Ekman and Friesen (1971) on this topic established the essential conceptual foundations for the development of measures in the field (Ekman, 1972, 1993; Ekman & Friesen, 1979). Among these contributions, the development of Pictures of Facial Affect (POFA; Ekman & Friesen, 1976)—a standardized set of black-and-white photographs of adult actors portraying prototypical expressions of basic emotions and neutral faces, widely used in experimental research—stands out.

Recent advances in FER measurement instruments include the diversification of stimulus banks, which now encompass different age groups (Conley et al., 2018; Donadon et al., 2019; Ebner et al., 2010; LoBue & Thrasher, 2014; Negrão et al., 2021) and ethnic and racial groups (Happy et al., 2017; Kim et al., 2017; Ma et al., 2020). These developments have also incorporated procedures aimed at enhancing ecological validity, such as the use

of colored stimuli and poses with the mouth open and closed (Tottenham et al., 2019). Furthermore, the operationalization of FER tasks has become increasingly sophisticated, involving computational resources for stimulus presentation, manipulation of exposure time and emotional expression intensity, and multimetric measurement approaches (e.g., accuracy, reaction time, and response bias) (Shimizu et al., 2025; Steephen et al., 2021; Tholl et al., 2024; Williams et al., 2023). Examples of tasks widely recognized and used in research settings include the Penn Emotion Recognition Test (Kohler et al., 2004), the Emotion Recognition Task (Montagne et al., 2007), and the Diagnostic Analysis of Nonverbal Accuracy–2nd Edition (DANVA-2; Nowicki & Carton, 1993).

Few resources are available for psychological and neuropsychological assessment in clinical practice, particularly in Brazil. According to data from the *SATEPSI – Sistema de Avaliação de Testes Psicológicos* (Psychological Test Assessment System), proposed by the Brazilian Federal Council of Psychology to evaluate, regulate, and disseminate instruments that can be legally used by psychologists in professional practice (CFP, 2022), there are currently no tests available for this specific purpose. This limitation restricts the application of such instruments to complementary and exploratory procedures and, consequently, narrows the scope of interpretation of results. Furthermore, the use of clinical tasks derived from experimental paradigms to explore this skill has been reported (Mioto et al., 2023); however, these tasks frequently lack adequate standardization and robust psychometric properties.

Therefore, developing a FER test adapted to the Brazilian context is warranted, as it would allow for the appropriate measurement of these skills using multiple performance metrics, such as accuracy and response time.

Based on the preceding discussion, this study aims to present the development of the Facial Emotion Recognition Task (TAREFE), a computerized test designed to evaluate facial expression recognition of emotions in the Brazilian adult population, with an emphasis on the methodological procedures adopted and the initial psychometric indicators. It is expected that, following additional robust psychometric validation, TAREFE will fill this critical gap, emerging as a culturally adapted and regulated instrument for clinical use in Brazil.

Method

Stimulus development

The images used in the TAREFE instrument were derived from a photographic database produced and validated by Rodrigues (2015). In that study, 15 actors were selected through convenience sampling. The group included Performing Arts students (men [n = 9] and women [n = 6]) with a mean age of 22.9 years (± 3.9), representing different ethnic and racial groups. The actors portrayed expressions corresponding to the emotions of joy, sadness, fear, anger, surprise, disgust, and neutrality. Prior to image acquisition, the actors read an informational text on basic emotional expressions and underwent training, during which they were instructed to pose the expressions as naturally as possible. For the emotions of anger, disgust, joy, and neutral, actors were also asked to produce distinct poses with their mouths open and closed.

All actors wore a black cape over their shoulders and, when necessary, used a hair clip. Image acquisition conditions were standardized, with a white background and an adjustable chair positioned 1.2 m from the camera. Images were captured using a Canon EOS Rebel T3i camera. Lighting was provided by a Nissin Mark II Di 866 Professional flash, ensuring homogeneity across images.

A total of 1,448 stimuli were generated, of which 170 were initially selected after a screening conducted by the researchers, based on emotional representativeness and technical quality criteria. The stimuli were coded for identification according to the following parameters: sex (female [F], male [M]), actor number (1 to 9), emotional expression (joy [AL], fear [ME], anger [RA], surprise [SU], sadness [TR], disgust [NO], and neutral [NE]), mouth position (open [BA] or closed [BF]), and photograph number, given that more than one image of a specific emotion was selected for some actors. Accordingly, the final code for each stimulus followed the format:

F_01_AL_BF_01. In this example, the stimulus represents female actor number 1 displaying a joyful expression with the mouth closed, corresponding to photograph number 1.

TAREFE development

TAREFE was designed to consist of stimuli depicting the emotions of joy, sadness, disgust, anger, fear, surprise, and neutral, with two male and two female faces for each emotion, regardless of mouth position (open or closed). In this study, the term "stimulus" refers to each individual item, that is, each photograph portraying an emotional facial expression. To select the most appropriate stimuli, Item Response Theory (IRT) was applied, based on data from Rodrigues (2015).

Sample and data collection

The sample consisted of 242 university students (78% women), aged between 18 and 40 years (mean age = 21.8 years). Stimuli were presented to participants on a 21.5-inch monitor using SuperLab software. Participants were asked to select the emotion that best represented the facial expression displayed. Accuracy rates were highest for the emotion of joy (94.07%) and lowest for fear (38.45%). Interrater agreement showed a Kappa coefficient greater than 0.70 for the vast majority of stimuli.

Data analysis

Descriptive statistics were initially used to assess the frequency of errors and correct answers for each stimulus and examine response variability. In addition, the assumptions required for Item Response Theory (IRT) analyses were evaluated, namely: a) unidimensionality, which assumes that items assess a single underlying construct. To address this assumption, the Bootstrap Modified Parallel Analysis Test (BMPAT; Finch & French, 2015) was employed. The BMPAT was selected due to its greater robustness for investigating dimensionality in small samples and categorical data, minimizing instability problems frequently observed in traditional factor analyses. In accordance with previous studies in the field (Schlegel et al., 2014), items representing the same emotion were analyzed together, resulting in seven emotion-specific analyses; b) local independence, according to which an individual's response to one item should not influence responses to other items, with this independence conditional exclusively on the latent trait. This assumption was evaluated using Yen's Q3 index (1984), with the bias correction proposed by Yen (1993). Q3 values > 0.20 were considered indicative of local dependence (Chen & Thissen, 1997). The assumption was considered adequately met when fewer than 20% of item pairs presented values above this criterion (de Ayala, 2009; Marais & Andrich, 2008); c) monotonicity, which assumes that individuals with higher levels of the latent trait present a greater probability of endorsing or correctly answering the items. This assumption was investigated using the check monotonicity function from the Mokken package (R), which estimates conditional item response probabilities based on groups formed according to participants' rest scores, verifying whether the probability of a correct response increases monotonically along the latent trait continuum. The number of monotonicity violations, the number of valid comparisons, and the statistical significance of violations were evaluated (Finch & French, 2015). Model specification assumed a logistic or normal distribution of errors. More detailed information regarding these analyses is available in Supplementary Material (SM1).

Subsequently, analyses were conducted using the Rasch Model (one-parameter logistic model; 1PL) to identify stimuli exhibiting misfit based on infit and outfit indices. Infit assesses unexpected response patterns among individuals whose latent trait level (theta) is close to the item's difficulty, reflecting variation within the expected response range. Outfit, in contrast, assesses unexpected response patterns among individuals whose theta levels are substantially below or above the item's difficulty, capturing atypical responses that deviate from the expected pattern (Boone et al., 2013). For samples of up to 500 respondents, acceptable reference values for infit and outfit indices range from 0.7 to 1.3 (Bond et al., 2020; Boone, 2016). Standardized fit statistics (infit t and outfit t) were

also examined to assess model adequacy by identifying potential discrepancies. According to the framework proposed by Linacre (2002), values between −2 and 2 indicate acceptable model fit, with lower values suggesting overly predictable data and higher values indicating excessive unpredictability.

Concurrently, analyses were conducted using the two-parameter logistic (2PL) model to investigate the discrimination and difficulty parameters of each stimulus. As in the previous analysis, potential inadequacies in infit and outfit indices were evaluated using the same reference criteria. Following the methodological considerations regarding the use of MNSQ statistics in a 2PL framework, an additional item-fit analysis was conducted using Orlando and Thissen's S-X^2 statistic and its associated p-value and RMSEA, which are specifically applicable to evaluating item fit in the 2PL model. The S-X^2 results were interpreted based on the statistical significance of the test and the magnitude of the associated RMSEA. The S-X^2 analysis provided an additional 2PL-specific assessment of item fit (Orlando & Thissen, 2003). In addition, item difficulty (*b*) and discrimination (*a*) parameters were estimated for each stimulus. The classification criteria proposed by Baker and Kim (2017) were adopted: a) for difficulty (*b*): stimuli with values <−2.62 were classified as "very easy"; values between −2.62 and −1.50 as "easy"; between −1.5 and 1.5, as "moderate"; between 1.50 and 2.62 as "difficult"; and values above 2.62 as "very difficult"; and b) for discrimination (a), values equal to 0 indicated no discriminatory power; values between 0.1 and 0.34 indicated very low discrimination; between 0.35 and 0.64, low discrimination; between 0.65 and 1.34, moderate discrimination; between 1.35 and 1.69, high discrimination; and values above 1.70 indicated very high discrimination. Item Characteristic Curves (ICCs), which graphically illustrate item difficulty and discrimination, were also analyzed (Andrade et al., 1998).

Initial evidence of TAREFE reliability (Stage 1) was assessed using Omega coefficient (Baker & Kim, 2018; McDonald, 1999). The analyses were conducted in the RStudio® interface (v.2023.12.1; R Core Team, v.4.3.3). The *readxl* package was used to import.xlsx (Excel) files; the *TAM* package was employed to estimate the 2PL IRT model and obtain fit indices (infit and outfit); the *ltm* package was used to estimate the same model and generate ICCs; the psych package was use to estimate Omega coefficient. Parameter estimation was performed using the Marginal Maximum Likelihood (MML) method, which estimates item parameters only (Andrade et al., 2000).

Task operationalization

The selection of stimuli comprising the final instrument followed these main criteria: a) an equal number of stimuli for each emotion, totaling four per emotion—two male and two female faces; b) acceptable item fit across both the 1PL and 2PL IRT models, based on the joint examination of Infit and Outfit MNSQ statistics, with items showing problematic fit under either model excluded from further consideration; the retained items were subsequently evaluated under the 2PL model based on their S-X^2 analysis, item-specific difficulty and discrimination parameters. When necessary, borderline fit values were accepted in order to satisfy the remaining predefined additional criteria: a) preference for stimuli with greater discriminatory power; b) preference for items with at least moderate difficulty, given that most stimuli were classified as easy or very easy; c) maintenance of balance in the number of stimuli per actor; and d) visual inspection of the ICCs.

The TAREFE will developed and implemented using the selected stimuli. This task is designed to assess participants' ability to recognize facial expressions of emotion. Participants are required to analyze a facial stimulus and answer the question: *"What emotion is expressed in this face?"* by selecting one of the seven possible response alternatives—joy, sadness, fear, disgust, surprise, anger, and neutral—presented in random order to minimize response bias. Only one response is correct. Responses are scored dichotomously, with 0 indicating an incorrect response and 1 indicating a correct response. Although response time is not restricted, it is automatically recorded by the software for each stimulus, corresponding to the interval between stimulus onset and response selection. The pattern of errors is also recorded by the software.

Table 1 Item fit indicators for different emotional stimuli under the one-parameter logistic IRT (Rasch) model

Stimulus	*Infit*	*Infit_t*	*Outfit*	*Outfit_t*
Joy				
F_01_AL_BF_01	.96	-.08	.78	-.74
F_02_AL_BA_01	1.01	.12	.93	-.16
F_02_ AL_BF_01	1.01	0.09	1.06	.36
F_05_AL_BA_01	.95	-.07	**.68**	-.89
F_05_AL_BF_01	1.01	.13	1.29	1.01
F_06_AL_BA_01	1.02	.18	**1.45**	1.06
F_06_AL_BF_01	1.03	.99	1.02	.48
M_01_AL_BF_01	.96	-.02	.79	.49
M_02_AL_BF_01	1.03	.22	10.05	.30
M_03_AL_BA_01	.99	.07	.92	-.08
M_03_AL_ BF_01	1.05	.50	1.09	.76
M_04_AL_BA_01	.96	-.01	0.87	-.25
M_04_ AL_BF_01	.96	-.22	0.91	-.51
M_05_AL_BA_01	.97	.04	1.18	.54
M_05_AL_BF_01	.97	-.12	0.89	-.39
M_06_AL_BA_01	.99	.10	0.77	-.38
M_06_AL_BF_01	.96	-.07	0.71	-.92
M_07_AL_BA_01	1.02	.16	1.03	0,20
M_08_AL_BA_01	.98	.09	**0.64**	-.73
M_08_ AL_BF_01	.95	-.05	**0.60**	−1.02
M_09_AL_BA_01	.95	-.04	0.93	-.02
M_09_AL_BF_01	94	-.44	0.79	−1.38
Fear				
F_01_ME_BA_01	.93	−1.85	.92	−1.58
F_02_ME_BA_01	.98	-.26	.98	-.28
F_03_ME_BA_01	.92	**−2.4**	.91	**−2.19**
F_04_ME_BA_01	.92	**−2.82**	.91	**−2.62**
F_05_ME_BA_01	.91	−1.48	.86	−1.74
F_06 _ME_BA_01	.9	**−2.09**	.89	−1.83
M_01_ME_BA_01	.93	−1.9	.92	−1.74
M_02_ME_BA_01	.88	**−3.5**	.86	**−3.44**
M_03_ME_BA_01	.89	**−3.26**	.88	**−2.95**
M_04_ME_BA_01	.87	**−3.51**	.84	**−3.6**
M_05_ME_BA_01	.91	**−3.08**	.9	**−2.78**
M_07_ME_BA_01	.94	−1.19	.92	−1.28
M_08_ME_BA_01	.9	**−2.83**	.89	**−2.4**
M_09_ME_BA_01	.95	-.99	.95	-.87
Neutral				
F_01_NE_BA_01	.99	-.04	.95	-.41
F_01_NE_BF_01	1.00	.05	.81	-.67
F_02_NE_BA_01	.98	.01	.73	-.94
F_02_NE_BF_01	1.05	.31	1.24	.94
F_03_NE_BA_01	1.03	.33	.99	-.04
F_03_NE_BF_01	1.01	.13	1.00	.13
F_04_NE_BA_01	1.00	.04	.87	-.52
F_05_NE_BA_01	.96	-.22	.89	-.59
F_05_NE_BF_01	.96	-.16	.74	−1.17
F_06_NE_BA_01	1.04	1.10	1.04	1.06
F_06_NE_BF_01	1.06	1.89	1.07	1.80
M_01_NE_BA_01	.97	-.52	.95	-.63
M_01_NE_BF_01	1.00	.07	.96	-.04
M_02_NE_BA_01	.97	-.14	.88	-.57
M_02_NE_BF_01	.99	.10	.77	-.43

Table 1 (continued)

Stimulus	*Infit*	*Infit_t*	*Outfit*	*Outfit_t*
M_03_NE_BA_01	1.01	.09	.97	-.26
M_03_NE_BF_01	1.01	.13	.91	-.18
M_04_NE_BA_01	1.02	.40	1.00	.07
M_04_NE_BF_01	1.01	.08	.96	-.16
M_05_NE_BA_01	.99	-.41	.99	-.39
M_05_NE_BF_01	.97	.05	**.58**	-.90
M_06_NE_BA_01	.98	-.02	.90	-.29
M_06_NE_BF_01	.99	.07	.71	-.66
M_07_NE_BA_01	.99	.01	.83	-.63
M_07_NE_BF_01	.99	.03	.88	-.32
M_08_NE_BA_01	.98	.01	.82	-.54
M_09_NE_BA_01	.93	-.41	.75	−1.28
Anger				
F_01_RA_BF_01	.95	-.10	.75	-.72
F_02_RA_BA_01	1.00	.10	.75	-.55
F_03_RA_BF_01	.92	-.25	**.67**	−1.18
F_04_RA_BA_01	.98	-.28	.97	.31
F_04_RA_BF_01	.98	-.14	.95	-.39
F_05_RA_BA_01	1.01	.13	.92	-.16
F_05_RA_BF_01	1.02	.17	.96	-.03
F_06_RA_BA_01	1.00	.06	.90	-.25
F_06_RA_BF_01	.97	-.01	.98	.07
M_01_RA_BF_01	.99	.06	.91	-.13
M_02_RA_BA_01	1.11	**2.29**	1.20	**3.24**
M_02_RA_BF_01	1.07	1.84	1.07	1.63
M_03_RA_BA_01	1.06	.73	1.08	.86
M_03_RA_BF_01	1.01	.16	.97	.08
M_04_RA_BF_01	1.01	.13	.99	.10
M_05_RA_BA_01	.99	.02	.84	-.66
M_05_RA_BF_01	1.03	.65	1.03	.57
M_06_RA_BA_01	1.01	.09	.95	-.22
M_07_RA_BF_01	.95	-.03	**.69**	-.74
M_08_RA_BA_01	.95	-.07	.86	-.26
M_08_RA_BF_01	.96	-.11	.77	-.97
M_09_RA_ BA_01	1.02	.26	1.02	.25
M_09_RA_BF_01	.95	-.03	**.58**	−1.02
Surprise				
F_01_SU_BA_01	1.08	.57	**1.37**	1.92
F_02_SU_BA_01	1.15	1.16	**1.82**	**4.14**
F_03_SU_BA_01	1.00	.09	1.32	1.15
F_03_SU_BA_02	1.00	.08	.99	-.15
F_04_SU_BA_01	.96	-.04	1.01	.16
F_05_SU_BA_01	.99	.05	.94	-.11
F_06_SU_BA_01	1.06	1.85	1.07	1.82
M_01_SU_BA_01	1.16	**2.32**	1.29	**3.25**
M_02_SU_BA_01	.99	-.06	1.05	.32
M_03_SU_BA_01	1.19	**2.88**	**1.34**	**3.91**
M_04_SU_BA_01	.98	-.11	.96	-.22
M_05_SU_BA_01	1.09	.73	**1.31**	1.87
M_06_SU_BA_01	.99	.01	1.01	.11
M_07_SU_BA_01	.98	-.09	.98	.06
M_08_SU_BA_01	1.01	.14	1.06	.29
M_08_SU_BA_02	1.01	.27	1.02	.36
M_09_SU_BA_01	.99	-.08	.95	-.38

Table 1 (continued)

Stimulus	*Infit*	*Infit_t*	*Outfit*	*Outfit_t*
Sadness				
F_01_TR_BF_01	.97	-.01	.80	-.43
F_02_TR_BF_01	.97	-.06	.85	-.47
F_03_TR_BF_01	1.00	.07	.93	-.40
F_04_TR_BF_01	.99	-.02	1.03	.23
F_05_TR_BF_01	1.00	.07	.89	-.31
F_06_TR_BF_01	1.10	**3.04**	1.12	**2.98**
M_01_TR_BF_01	.97	-.17	.88	-.64
M_02_TR_BF_01	1.06	1.52	1.11	**2.29**
M_03_TR_BF_01	.97	-.87	.98	-.41
M_04_TR_BF_01	1.01	.14	.91	-.22
M_05_TR_BF_01	1.04	.48	1.09	.86
M_05_TR_BF_02	.99	.02	1.05	.31
M_06_TR_BF_01	1.07	1.14	1.12	1.66
M_07_TR_BF_01	.97	-.03	.88	-.33
M_08_TR_BF_01	.97	-.14	.93	-.37
M_09_TR_BF_01	.97	-.17	.94	-.34
M_09_TR_BF_02	.96	-.30	.90	-.59
M_09_TR_BF_03	.96	-.28	.95	-.23
Disgust				
F_02_NO_BA_01	.96	-.06	.80	-.52
F_02_NO_BF_01	.98	-.12	.86	-.74
F_03_NO_BA_01	.98	.00	.90	-.20
F_03_NO_BF_01	1.03	.22	1.09	.56
F_04_NO_BA_01	.99	.10	.81	-.27
F_04_NO_BF_01	.98	.02	.82	-.51
F_05_NO_BF_01	1.06	1.10	1.12	1.57
F_06_NO_BA_01	1.07	1.28	1.12	1.62
F_06_NO_BF_01	1.05	1.51	1.09	**2.20**
M_01_NO_BA_01	1.08	**2.50**	1.12	**2.86**
M_01_NO_BF_01	1.10	1.96	1.13	1.99
M_02_NO_BA_01	.99	.09	.80	-.37
M_02_NO_BF_01	1.01	.14	1.08	.43
M_03_NO_BA_01	.98	.09	**.68**	-.62
M_03_NO_BA_02	1.08	**2.51**	1.12	**2.89**
M_04_NO_BA_01	1.04	1.22	1.08	**2.04**
M_04_NO_BF_01	1.06	1.21	1.13	**2.07**
M_05_NO_BA_01	.97	-.13	.95	-.17
M_05_NO_BF_01	1.05	1.65	1.06	1.45
M_06_NO_BA_01	1.12	**2.92**	1.17	**3.22**
M_06_NO_BF_01	1.07	**2.18**	1.09	**2.40**
M_07_NO_BA_01	.98	.07	.77	-.44
M_07_NO_BF_01	1.07	**2.26**	1.10	**2.37**
M_08_NO_BA_01	.99	-.13	.91	-.80
M_08_NO_BF_01	1.10	1.72	1.14	1.98
M_09_NO_BA_01	.95	.00	.74	-.45
M_09_NO_BA_02	.95	-.06	.74	-.65
M_09_NO_BF_01	1.01	.09	.98	-.04

Abbreviations: AL=Joy; BA=open mouth; BF=closed mouth; F=Female; M=Male; ME=Fear; NE=Neutral; NO=Disgust; RA=Anger; SU=Surprise; t=standardized value; TR=Sadness. **Boldface** indicates values outside expected parameters

Results

Stimulus Selection

Based on the analysis of response frequencies, 21 stimuli (1.8% of the total) were excluded because they did not exhibit sufficient variability for subsequent analyses (number of errors≤5). These stimuli were: F_01_AL_BA_01, F_01_RA_BA_01, F_02_NO_BA_02, F_02_RA_BF_01, F_03_AL_BA_01, F_03_AL_BF_01, F_03_RA_BA_01, F_03_TR_BF_02, F_04_AL_BA_01, F_04_AL_BF_01, F_04_NE_BF_01, F_05_NO_BA_01, M_01_AL_BA_01, M_01_RA_BA_01, M_02_AL_BA_01, M_04_RA_BA_01, M_06_RA_BF_01, M_07_AL_BF_01, M_07_RA_BA_01, M_08_NE_BF_01, and M_09_NE_BF_01. As a result, 149 photographs remained for evaluation using logistic models. The joy category comprised 22 stimuli (seven with female faces), fear included 14 stimuli (six with female faces), neutral comprised 27 stimuli (11 with female faces), anger included 23 stimuli (nine with female faces), surprise included 17 stimuli (seven with female faces), sadness comprised 18 stimuli (six with female faces), and disgust comprised 28 stimuli (nine with female faces). Table 1 presents the results of the 1PL IRT model for these stimuli.

According to Table 1, based on the Rasch Model (1PL) analysis, one stimulus exhibited misfit exclusively on the infit index, whereas 16 stimuli showed misfit only on the outfit index. An additional 16 stimuli exhibited misfit on both indices, resulting in a total of 33 stimuli with inadequate fit. Table 2 presents the results of the 2PL IRT analysis.

As shown in Table 2, according to the 2PL IRT model, none of the stimuli exhibited misfit on the infit index; however, 14 stimuli showed misfit on the outfit index and six other items showed a misfit in the S-X^2 analysis. In addition, two stimuli associated with the emotion of surprise showed no discriminatory power; nine stimuli were classified as having very low discrimination, 14 as low, 62 as moderate, 32 as high, and 30 as very high discrimination. With respect to the difficulty parameter, 56 stimuli were classified as very easy, 43 as easy, 47 as having moderate difficulty, one as difficult, and two as very difficult.

The Item Characteristic Curves (ICCs) for each stimulus are presented in the supplementary material (SM2) for further analysis. Based on the previously defined criteria, the stimulus selection process for each emotion is described below:

- Joy – female faces: of the seven available stimuli, three were excluded due to outfit misfit (F_01_AL_BF_01; F_05_AL_BA_01; F_06_AL_BA_01). Stimulus F_06_AL_BF_01 was also excluded due to its low level of discrimination. Among the remaining stimuli, two depicted the same actress (F_02_AL_BA_01 and F_02_AL_BF_01); F_02_AL_BF_01 was selected because it showed a higher level of discrimination (very high) and easy difficulty. This stimulus was combined with F_05_AL_BF_01, which exhibited moderate discrimination, very easy difficulty, and satisfactory ICCs.
- Joy – male faces: of the 15 available stimuli, six showed outfit misfit and were excluded (M_01_AL_BF_01, M_03_AL_BA_01, M_05_AL_BA_01, M_07_AL_BA_01, M_08_AL_BA_01, and M_08_AL_BF_01). No stimuli were excluded based on the discrimination parameter, as all showed moderate to very high discriminatory power. Regarding difficulty, three were classified as very easy and were excluded (M_04_AL_BA_01, M_06_AL_BA_01, M_09_AL_BA_01). Among the remaining stimuli (M_02_AL_BF_01, M_03_AL_BF_01, M_04_AL_BF_01, M_05_AL_BF_01, M_06_AL_BF_01, M_09_AL_BF_01), the highest difficulty levels were observed for M_03_AL_BF_01 and M_09_AL_BF_01 (moderate difficulty). These stimuli would have been preferred; however, after applying additional criteria, stimulus M_09_AL_BF_01 was excluded in favor of M_04_AL_BF_01 (easy difficulty, high discriminatory power) to ensure balance among actors.
- Fear—female faces: of the six available stimuli, three (F_03_ME_BA_01, F_04_ME_BA_01, and F_06_ME_BA_01) were excluded due to misfit on infit and/or outfit indices. Among the remaining stimuli, F_05_ME_BA_01 was excluded because it presented a standardized outfit value at the decision threshold (−2.00), leaving stimuli F_01_ME_BA_01 and F_02_ME_BA_01, which were selected.

Table 2 Fit indicators, discrimination, and difficulty levels of different emotional stimuli under the two-parameter logistic IRT model

TRI-2PL										
Stimulus	***Infit***	***Infit_t***	***Outfit***	***Outfit_t***	**RMSEA** $_{s\text{-}x}^{2}$	$p_{s\text{-}x}^{2}$	*a*	***CI***	***b***	***CI***
Joy										
F_01_AL_BF_01	1.03	.19	**.58**	**−2.66**	.000	.475	2.96	VH	−1.89	E
F_02_AL_BA_01	1.01	.10	.95	-.47	.000	.838	1.50	H	−2.55	E
F_02_AL_BF_01	.98	-.10	1.11	.30	.000	.496	1.90	VH	−1.75	E
F_05_AL_BA_01	.98	.02	1.07	.21	.000	.974	1.10	Md	−3.46	VE
F_05_AL_BF_01	1.02	.15	.82	-.88	.042	.167	1.34	Md	−2.73	VE
F_06_AL_BA_01	1.00	.13	.98	.00	.027	.301	.79	Md	−5.01	VE
F_06_AL_BF_01	1.00	.04	1.00	.04	.000	.710	.24	VL	−1.42	Md
M_01_AL_BF_01	.99	.03	**3.50**	.16	.000	.595	2.74	VH	−2.21	E
M_02_AL_BF_01	1.00	.04	.97	-.34	.000	.885	1.46	H	−2.13	E
M_03_AL_BA_01	1.05	.27	**.63**	−1.45	.052	.175	1.52	H	−2.97	VE
M_03_AL_BF_01	1.01	.11	.98	-.34	.000	.902	1.49	H	−1.38	Md
M_04_AL_BA_01	.98	.04	1.17	.39	.000	.513	1.13	Md	−3.50	VE
M_04_AL_BF_01	1.00	.00	1.02	-.16	.000	.726	1.66	H	−1.72	E
M_05_AL_BA_01	.97	-.01	**1.83**	.54	.000	.501	2.13	VH	−2.57	E
M_05_AL_BF_01	1.03	.22	.90	−1.13	.000	.527	1.99	VH	−1.97	E
M_06_AL_BA_01	1.00	.14	.96	-.03	.061	.133	.84	Md	−4.94	VE
M_06_AL_BF_01	1.00	.04	1.26	-.16	.000	.661	2.31	VH	−2.18	E
M_07_AL_BA_01	1.04	.24	**.51**	**−2.48**	.050	.169	2.31	VH	−2.28	E
M_08_AL_BA_01	1.02	.18	**.55**	−1.50	.000	.482	1.68	H	−2.99	VE
M_08_AL_BF_01	1.02	.16	**.57**	**−2.02**	.012	.308	2.41	VH	−2.37	E
M_09_AL_BA_01	.99	.07	1.16	-.29	**.101**	.051	1.66	H	−2.91	VE
M_09_AL_BF_01	1.01	.09	.94	−1.04	.000	.714	2.12	VH	−1.48	Md
Fear										
F_01_ME_BA_01	.99	-.13	1.08	.96	.000	.983	1.65	H	-.41	Md
F_02_ME_BA_01	1.01	.20	.94	-.94	.000	.539	1.07	Md	1.12	Md
F_03_ME_BA_01	1.00	-.01	1.03	.32	.047	.098	1.78	VH	.23	Md
F_04_ME_BA_01	1.00	.07	.94	-.84	.000	.807	2.09	VH	.03	Md
F_05_ME_BA_01	1.02	.23	.85	−2.00	.000	.539	2.00	VH	.82	Md
F_06_ME_BA_01	1.01	.10	.95	-.82	.037	.196	1.62	H	.62	Md
M_01_ME_BA_01	1.01	.15	.89	−1.62	.000	.974	1.91	VH	.39	Md
M_02_ME_BA_01	1.01	.14	.98	-.28	.000	.510	1.27	Md	-.30	Md
M_03_ME_BA_01	1.01	.20	.90	−1.46	.059	**.046**	2.08	VH	.25	Md
M_04_ME_BA_01	1.00	.01	1.02	.26	.040	.163	1.56	H	-.35	Md
M_05_ME_BA_01	1.00	.04	.94	-.80	.000	.741	2.19	VH	.10	Md
M_07_ME_BA_01	1.02	.30	.90	−1.51	.034	.213	1.45	H	.73	Md
M_08_ME_BA_01	1.00	-.01	.94	-.90	.000	.794	2.09	VH	.37	Md
M_09_ME_BA_01	1.01	.19	.97	-.55	.000	.726	1.01	Md	.84	Md
Neutral										
F_01_NE_BA_01	.99	-.05	1.00	-.06	.000	.970	1.48	H	−1.39	Md
F_01_NE_BF_01	1.00	.05	1.05	.26	.000	.747	.82	Md	−3.76	VE
F_02_NE_BA_01	.92	-.27	**1.37**	1.10	.047	.167	1.98	VH	−2.23	E
F_02_NE_BF_01	1.00	.08	1.00	.06	.034	.230	.18	VL	−14.85	VE
F_03_NE_BA_01	1.00	.05	1.01	-.29	.000	.527	2.11	VH	−1.03	Md
F_03_NE_BF_01	.98	.02	1.23	.74	.000	.945	1.29	Md	−3.03	VE
F_04_NE_BA_01	.97	-.12	1.17	.56	.000	.823	1.86	VH	−1.98	E
F_05_NE_BA_01	.98	-.10	1.18	1.12	.000	.680	1.37	H	−1.96	E
F_05_NE_BF_01	1.03	.24	.89	-.62	.000	.865	1.09	Md	−2.73	VE
F_06_NE_BA_01	.99	-.08	.97	-.47	.059	**.046**	1.36	H	-.27	Md
F_06_NE_BF_01	.99	-.13	1.01	.17	.018	.370	.81	Md	.13	Md
M_01_NE_BA_01	1.00	.04	1.02	.15	.000	.803	1.42	H	-.85	Md

Table 2 (continued)

TRI-2PL										
M_01_NE_BF_01	1.00	.07	1.03	.16	.000	.691	1.04	Md	−3.34	VE
M_02_NE_BA_01	1.00	.06	1.03	-.25	.000	.596	2.22	VH	−1.67	E
M_02_NE_BF_01	1.03	.19	.74	-.77	.000	.927	1.15	Md	−3.71	VE
M_03_NE_BA_01	1.00	-.01	1.09	.75	.000	.465	1.68	H	−1.14	Md
M_03_NE_BF_01	1.05	.26	.83	-.76	.059	.080	1.40	H	−2.80	VE
M_04_NE_BA_01	1.00	-.05	1.01	.02	.000	.762	1.54	H	-.78	M
M_04_NE_BF_01	.99	-.02	1.06	.35	.000	.741	1.29	Md	−2.00	E
M_05_NE_BA_01	1.00	.08	.97	-.54	.000	.680	1.40	H	.08	M
M_05_NE_BF_01	1.02	.18	1.02	.06	.056	.174	1.05	Md	−4.12	VE
M_06_NE_BA_01	1.03	.20	.96	-.26	.000	.765	1.12	Md	−2.96	VE
M_06_NE_BF_01	.97	.02	1.05	.21	.050	.165	1.07	Md	−3.77	VE
M_07_NE_BA_01	1.09	.50	**.68**	−1.81	.075	**.012**	1.49	H	−2.36	E
M_07_NE_BF_01	.95	-.16	**1.50**	1.53	.000	.675	1.63	H	−2.45	E
M_08_NE_BA_01	.92	-.27	**1.34**	.90	.057	.110	2.06	VH	−2.19	E
M_09_NE_BA_01	1.02	.14	.83	−1.18	.028	.277	1.54	H	−1.99	E
Anger										
F_01_RA_BF_01	1.02	.15	.99	-.39	.000	.626	1.48	H	−2.76	VE
F_02_RA_BA_01	1.01	.12	1.02	.04	.000	.665	1.17	Md	−3.51	VE
F_03_RA_BF_01	1.00	.04	1.09	-.38	.000	.686	1.95	VH	−2.22	E
F_04_RA_BA_01	1.00	.06	.97	-.44	.000	.965	1.80	VH	-.86	Md
F_04_RA_BF_01	1.00	.03	.97	-.31	.000	.951	1.02	Md	−1.84	Easy
F_05_RA_BA_01	.99	.03	.96	-.20	.036	.235	1.18	Md	−3.12	VE
F_05_RA_BF_01	1.03	.19	.86	-.59	.015	.382	1.03	Md	−3.35	VE
F_06_RA_BA_01	1.01	.11	.94	-.24	.000	.849	.92	Md	−3.59	VE
F_06_RA_BF_01	1.00	.09	.99	.06	.014	.385	.50	L	−6.82	VE
M_01_RA_BF_01	1.01	.12	.94	-.17	.030	.285	.91	Md	−4.01	VE
M_02_RA_BA_01	1.00	.01	1.00	.07	.000	.542	.31	VL	2.05	D
M_02_RA_BF_01	1.00	.12	1.00	.14	.020	.352	.57	L	.61	Md
M_03_RA_BA_01	1.00	-.02	1.00	-.04	.010	.406	.63	L	−2.06	E
M_03_RA_BF_01	1.01	.15	.88	-.25	.000	.456	.87	Md	−4.79	VE
M_04_RA_BF_01	.99	.05	1.04	.18	.000	.728	.84	Md	−4.29	VE
M_05_RA_BA_01	.99	.03	1.01	.08	.020	.350	.69	Md	−3.91	VE
M_05_RA_BF_01	1.00	-.09	1.00	-.03	.050	.122	.69	Md	−1.04	Md
M_06_RA_BA_01	1.00	.06	1.00	.01	.059	.069	.52	L	−4.51	VE
M_07_RA_BF_01	1.02	.16	.76	-.99	.070	.072	1.46	H	−3.04	VE
M_08_RA_BA_01	.95	-.12	**2.71**	.99	.061	.153	2.51	VH	−2.27	E
M_08_RA_BF_01	.98	-.03	1.09	.30	.000	.433	1.28	Md	−2.53	E
M_09_RA_BA_01	.99	-.11	1.01	.06	.000	.970	1.14	Md	−1.54	E
M_09_RA_BF_01	1.00	.08	**.69**	−1.42	.000	.590	2.28	VH	−2.49	E
Surprise										
F_01_SU_BA_01	1.00	.05	1.01	.11	.000	.626	.23	VL	−9.20	VE
F_02_SU_BA_01	1.00	.06	1.00	.00	.048	.133	-.43	-	4.68	VD
F_03_SU_BA_01	1.00	.07	1.00	.08	.021	.348	.36	L	−7.88	VE
F_03_SU_BA_02	.99	-.08	.99	-.18	**.119**	**.000**	1.53	H	-.78	Md
F_04_SU_BA_01	1.00	.10	1.01	.12	.044	.203	.28	VL	−12.07	VE
F_05_SU_BA_01	1.00	.08	1.08	.21	.000	.745	1.27	Md	−2.76	VE
F_06_SU_BA_01	1.00	-.09	1.01	.22	.053	.115	.82	Md	.20	Md
M_01_SU_BA_01	1.00	.02	1.00	.06	.040	.183	.16	VL	−6.42	VE
M_02_SU_BA_01	1.02	.18	.95	-.39	.018	.369	.99	Md	−2.33	E
M_03_SU_BA_01	1.00	.02	1.00	.04	.000	.801	-.19	-	5.14	VD
M_04_SU_BA_01	.99	-.04	1.04	.16	.064	.060	1.45	H	−1.70	E
M_05_SU_BA_01	1.00	.04	1.00	.05	.000	.608	.20	VL	−9.28	VE
M_06_SU_BA_01	.99	-.01	1.08	.15	.000	.735	1.66	H	−2.31	E
M_07_SU_BA_01	1.00	.05	.97	-.34	.016	.377	1.63	H	−1.70	E

https://doi.org/10.1186/s41155-026-00410-4

Table 2 (continued)

TRI-2PL										
M_08_SU_BA_01	1.00	.12	.99	.07	.000	.452	.50	L	−7.22	VE
M_08_SU_BA_02	.99	-.13	.93	−1.00	.053	.116	2.31	VH	-.63	Md
M_09_SU_BA_01	1.02	.20	.80	−2.05	.047	.152	1.93	VH	−1.29	Md
Sadness										
F_01_TR_BF_01	.96	-.03	1.16	.43	.031	.280	1.19	Md	−3.38	VE
F_02_TR_BF_01	.99	.01	1.08	.35	.000	.927	1.05	Md	−3.17	VE
F_03_TR_BF_01	1.00	.06	1.06	.35	.051	.100	1.26	Md	−1.96	E
F_04_TR_BF_01	1.01	.11	1.00	-.11	.000	.554	1.31	Md	−2.23	E
F_05_TR_BF_01	1.00	.09	.98	.00	.000	.701	.51	L	−6.03	VE
F_06_TR_BF_01	1.00	.02	1.00	.02	.078	**.010**	.31	VL	-.67	Md
M_01_TR_BF_01	1.00	.01	1.00	.01	.030	.268	1.03	Md	−2.33	E
M_02_TR_BF_01	.99	-.15	1.00	.06	.000	.595	.77	Md	-.67	Md
M_03_TR_BF_01	1.00	.07	1.00	.01	.017	.371	.62	L	-.15	Md
M_04_TR_BF_01	1.00	.07	1.08	.29	.022	.340	1.03	Md	−3.37	VE
M_05_TR_BF_01	1.00	.01	1.00	.06	.000	.934	.72	Md	−2.11	E
M_05_TR_BF_02	.99	-.02	1.03	.20	.034	.236	.76	Md	−3.26	VE
M_06_TR_BF_01	1.01	.12	.98	-.25	.042	.165	.74	Md	−1.30	Md
M_07_TR_BF_01	1.02	.16	.92	-.38	.037	.222	1.14	Md	−3.04	VE
M_08_TR_BF_01	1.02	.15	.93	-.59	.023	.333	1.33	Md	−1.96	E
M_09_TR_BF_01	1.01	.13	.98	-.16	.000	.845	1.05	Md	−2.19	E
M_09_TR_BF_02	.99	-.09	1.07	.20	.000	.695	1.64	H	−1.64	E
M_09_TR_BF_03	.99	-.07	.98	-.54	.000	.823	2.08	VH	−1.51	E
Disgust										
F_02_NO_BA_01	1.00	.08	1.03	.13	.051	.162	.97	Md	−3.69	VE
F_02_NO_BF_01	.99	.01	.99	-.02	.000	.861	.64	L	−3.53	VE
F_03_NO_BA_01	1.04	.23	**.66**	−1.71	.035	.252	1.88	VH	−2.37	E
F_03_NO_BF_01	1.01	.13	.97	-.21	.043	.132	.80	Md	−2.99	VE
F_04_NO_BA_01	1.01	.12	.75	−1.02	.000	.906	1.88	VH	−2.81	VE
F_04_NO_BF_01	.99	.05	1.09	.39	.000	.883	.71	Md	−4.67	VE
F_05_NO_BF_01	1.00	-.01	1.00	-.03	.029	.257	.39	L	−2.33	E
F_06_NO_BA_01	1.00	.05	1.00	.06	.000	.778	.75	Md	−1.26	Md
F_06_NO_BF_01	.99	-.10	1.03	.56	.000	.782	1.10	Md	.18	Md
M_01_NO_BA_01	1.00	-.04	1.00	.00	.000	.631	.48	L	-.55	Md
M_01_NO_BF_01	1.00	.01	1.00	.00	.000	.946	.36	L	−2.04	E
M_02_NO_BA_01	.99	.08	.83	-.56	.000	.842	1.49	H	−3.09	VE
M_02_NO_BF_01	1.00	.07	1.01	.08	.000	.933	.90	Md	−2.95	VE
M_03_NO_BA_01	1.00	.10	1.14	.24	.040	.236	1.42	H	−3.31	VE
M_03_NO_BA_02	1.00	-.03	1.00	.03	.000	.826	.31	VL	-.62	Md
M_04_NO_BA_01	1.00	-.04	1.01	.27	.040	.152	.84	Md	.30	Md
M_04_NO_BF_01	1.00	.03	.99	-.17	.017	.371	.94	Md	.90	Md
M_05_NO_BA_01	.98	-.10	1.15	.91	.056	.052	.85	Md	−3.05	VE
M_05_NO_BF_01	.99	-.17	.99	-.28	.000	.949	.81	Md	-.31	Md
M_06_NO_BA_01	1.00	.06	1.00	.00	.000	.740	.36	L	1.31	Md
M_06_NO_BF_01	1.00	-.10	1.00	-.15	.049	.074	.48	L	-.10	Md
M_07_NO_BA_01	1.01	.13	.82	-.73	.000	.905	1.59	H	−2.97	VE
M_07_NO_BF_01	1.00	.00	1.01	.13	.038	.176	.90	Md	-.27	Md
M_08_NO_BA_01	1.00	.00	1.00	-.04	.000	.730	.71	Md	−2.15	E
M_08_NO_BF_01	.99	-.09	1.02	.29	.000	.773	.78	Md	1.15	Md
M_09_NO_BA_01	.99	.11	1.06	.24	.000	.373	.72	Md	−5.60	VE

Table 2 (continued)

TRI-2PL										
M_09_NO_BA_02	1.02	.16	.75	-.90	.000	.774	1.25	Md	−3.25	VE
M_09_NO_BF_01	.99	.02	1.01	.07	.000	.897	.84	Md	−3.19	VE

Abbreviations: *a* discrimination parameter, AL = joy; *b* = difficulty parameter, BA = open mouth, BF = closed mouth, *Cl* Classification, *D* Difficult, *E* Easy, *F* Female, *H* High, *L* Low, *M* Male, *Md* Moderate, *ME* Fear, *NE* Neutral, *NO* Disgust, RA = Anger, SU = Surprise, t = standardized value, TR = Sadness, VD = Very Difficult, VE = Very Easy, VH = Very high; VL = Very Low. **Boldface** indicates indicators outside expected parameters

▪ Fear—male faces: of the eight available stimuli, five were excluded due to misfit on the infit and/or outfit indices (M_02_ME_BA_01, M_03_ME_BA_01, M_04_ME_BA_01, M_05_ME_BA_01, and M_08_ME_BA_01). Among the three remaining stimuli (M_01_ME_BA_01, M_07_ME_BA_01, and M_09_ME_BA_01), the two with the highest level of discriminatory power were selected (M_01_ME_BA_01 and M_07_ME_BA_01), whereas M_09_ME_BA_01 was excluded because it showed the lowest discriminatory power.

▪ Neutral—female faces: of the 11 available stimuli, F_02_NE_BA_01 was excluded due to outfit misfit, and F_02_NE_BF_01 was excluded due to very low discriminatory power. Stimuli classified as very easy (F_01_NE_BF_01, F_03_NE_BF_01, F_05_NE_BF_01) and easy (F_04_NE_BA_01 and F_05_NE_BA_01) were also excluded. Of the four remaining stimuli, F_06_NE_BF_01 was excluded because it showed moderate discriminatory power, which was the lowest among the remaining stimuli. Between F_06_NE_BA_01 and F_01_NE_BA_01, both of which showed high discrimination and moderate difficulty, F_01_NE_BA_01 was selected due to a more favorable difficulty value, better ICCs based on visual inspection, and improved balance among actors. This stimulus was paired with F_03_NE_BA_01, which exhibited very high discriminatory power and moderate difficulty.

▪ Neutral – male faces: of the 16 stimuli analyzed, four were excluded due to outfit misfit (M_05_NE_BF_01, M_07_NE_BA_01, M_07_NE_BF_01, and M_08_NE_BA_01). No stimuli were excluded based on discriminatory power. However, stimuli classified as very easy (M_01_NE_BF_01, M_02_NE_BF_01, M_03_NE_BF_01, M_06_NE_BA_01, and M_06_NE_BF_01) and easy (M_04_NE_BF_01, M_02_NE_BA_01, and M_09_NE_BA_01) were also excluded. Of the four remaining stimuli, three (M_01_NE_BA_01, M_03_NE_BA_01, and M_04_NE_BA_01) exhibited moderate difficulty, and one was classified as difficult (M_05_NE_BA_01). Despite this, M_05_NE_BA_01 was not selected to maintain balance among actors. Therefore, M_03_NE_BA_01 and M_04_NE_BA_01 were selected because they exhibited higher discriminatory power than M_01_NE_BA_01.

▪ Anger—female faces: of the nine stimuli analyzed, F_03_RA_BF_01 was initially excluded due to outfit misfit. Stimulus F_06_RA_BF_01 was excluded due to its low discriminatory power, as were stimuli classified as very easy (F_01_RA_BF_01, F_02_RA_BA_01, F_05_RA_BA_01, F_05_RA_BF_01, and F_06_RA_BA_01). The two remaining stimuli (F_04_RA_BA_01 and F_04_RA_BF_01) depicted the same actress, with F_04_RA_BA_01 being preferred due to its very high discriminatory power. Accordingly, to maintain balance among actors, F_03_RA_BF_01 was reinstated, as it presented adequate psychometric parameters, despite exhibiting an outfit value at the decision threshold (0.67).

▪ Anger—male faces: of the 14 available stimuli, four were excluded due to misfit on the infit and/or outfit indices (M_02_RA_BA_01, M_07_RA_BF_01, and M_08_RA_BA_01, and M_09_RA_BF_01). Three stimuli were excluded due to low discriminatory power (M_02_RA_BF_01, M_03_RA_BA_01, and M_06_RA_BA_01), and five additional stimuli were excluded due to low difficulty levels—classified as very easy (M_01_RA_BF_01, M_03_RA_BF_01, M_04_RA_BF_01, and M_05_RA_BA_01) or easy (M_08_RA_BF_01). Because item M_09_RA_BA_01 was at the boundary between easy and moderate difficulty and showed moderate discriminatory power, it was retained, together with M_05_RA_BF_01, which exhibited moderate discriminatory power and moderate difficulty.

▪ Surprise—female faces: of the seven available stimuli, two were excluded due to outfit misfit (F_01_SU_BA_01 and F_02_SU_BA_01), and two others were excluded due to very low (F_04_SU_BA_01) or low (F_03_SU_BA_01) discriminatory power. Of the three remaining stimuli, F_03_SU_BA_02 would have been preferred due to its high level of discriminatory power; however, it was excluded in order to ensure balance among actors. Accordingly, stimuli F_05_SU_BA_01 (very easy difficulty) and F_06_SU_BA_01 (moderate difficulty) were selected, both of which exhibited moderate discriminatory power.
▪ Surprise—male faces: of the 10 available stimuli, three were excluded due to misfit on infit and/or outfit indices (M_01_SU_BA_01, M_03_SU_BA_01, and M_05_SU_BA_01), and one due to low discriminatory power (M_08_SU_BA_01). Among the remaining stimuli, M_02_SU_BA_01, M_04_SU_BA_01, M_06_SU_BA_01, and M_07_SU_BA_01 were classified as easy, whereas M_08_SU_BA_02 and M_09_SU_BA_01 were classified as having moderate difficulty. These latter two stimuli were initially preferred, as they also exhibited very high discriminatory power; however, to ensure balance among actors, stimulus M_07_SU_BA_01 was selected in place of M_08_SU_BA_02. Among the stimuli classified as easy, M_07_SU_BA_01 showed the highest numerical values for difficulty and discriminatory power.
▪ Sadness—female faces: of the six available stimuli, F_06_TR_BF_01 was excluded due to misfit on the infit and outfit indices, and F_05_TR_BF_01 was excluded due to low discriminatory power. Of the four remaining stimuli, all of which exhibited moderate discriminatory power, two (F_01_TR_BF_01 and F_02_TR_BF_01) were excluded because they were classified as very easy. Thus, the final selection comprised F_03_TR_BF_01 and F_04_TR_BF_01, both of which exhibited easy difficulty.
▪ Sadness—male faces: of the 12 available stimuli, M_02_TR_BF_01 was excluded due to outfit misfit, whereas M_03_TR_BF_01 was excluded due to low discriminatory power. Stimuli classified as very easy (M_04_TR_BF_01, M_05_TR_BF_02, M_07_TR_BF_01) were also excluded. Of the seven remaining stimuli, three (M_09_TR_BF_01, M_09_TR_BF_02, and M_09_TR_BF_03) depicted the same actor. Based solely on discriminatory power and difficulty criteria, the preferred choice among these would have been M_09_TR_BF_02, which showed high discriminatory power despite low difficulty. Another stimulus that would have been selected under these criteria was M_06_TR_BF_01, which exhibited moderate discriminatory power and moderate difficulty. However, to ensure balance among actors, these two stimuli were not selected. Instead, stimuli M_01_TR_BF_01 and M_08_TR_BF_01 were chosen, both of which showed moderate discriminatory power despite being classified as easy. This decision also led to the exclusion of M_05_TR_BF_01.
▪ Disgust—female faces: among the nine stimuli analyzed, two (F_03_NO_BA_01, F_06_NO_BF_01) were initially excluded due to outfit misfit. Two additional stimuli (F_02_NO_BF_01 and F_05_NO_BF_01) were excluded because they exhibited low discriminatory power. Of the remaining five stimuli, F_04_NO_BA_01 was selected because it was the only stimulus showing very high discriminatory power. Among the four stimuli exhibiting moderate discriminatory power (F_02_NO_BA_01, F_03_NO_BF_01, F_04_NO_BF_01, and F_06_NO_BA_01), F_06_NO_BA_01 was selected because it exhibited the highest difficulty level (moderate).
▪ Disgust—male faces: of the 19 stimuli analyzed, eight were excluded due to misfit on the infit and/or outfit indices (M_01_NO_BA_01, M_03_NO_BA_01, M_03_NO_BA_02, M_04_NO_BA_01, M_04_NO_BF_01, M_06_NO_BA_01, M_06_NO_BF_01, and M_07_NO_BF_01). Stimulus M_01_NO_BF_01 was also excluded due to low discriminatory power. Seven stimuli were excluded due to low difficulty—classified as very easy (M_02_NO_BA_01, M_02_NO_BF_01, M_05_NO_BA_01, M_07_NO_BA_01, M_09_NO_BA_01, M_09_NO_BA_02, and M_09_NO_BF_01)—whereas M_08_NO_BA_01 was classified as easy. Accordingly, M_05_NO_BF_01 and M_08_NO_BF_01 remained, both exhibiting moderate discriminatory power and moderate difficulty.

Table 3 presents the stimuli selected through these analyses, which comprise the TAREFE.

Table 3 Stimuli selected for the composition of the TAREFE instrument

Emotions	Stimulus					
	Female Faces	***a***	***b***	**Male Faces**	***a***	***b***
Joy	F2_AL_BF_01	VH	E	M3_AL_BF_01	H	Md
	F5_AL_BF_01	Md	VE	M4_AL_BF_01	H	E
Fear	F1_ME_BA_01	H	Md	M1_ME_BA_01	VH	Md
	F2_ME_BA_01	Md	Md	M7_ME_BA_01	H	Md
Neutral	F1_NE_BA_01	H	Md	M3_NE_BA_01	H	Md
	F3_NE_BA_01	VH	Md	M4_NE_BA_01	H	Md
Anger	F3_RA_BF_01	VH	E	M5_RA_BF_01	Md	Md
	F4_RA_BA_01	VH	Md	M9_RA_BA_01	Md	E
Surprise	F5_SU_BA_01	Md	VE	M7_SU_BA_01	H	E
	F6_SU_BA_01	Md	Md	M9_SU_BA_01	VH	Md
Sadness	F3_TR_BF_01	Md	E	M1_TR_BF_01	Md	E
	F4_TR_BF_01	Md	E	M8_TR_BF_01	Md	E
Disgust	F3_NO_BA_01	VH	VE	M5_NO_BF_01	Md	Md
	F6_NO_BA_01	Md	Md	M8_NO_BF_01	Md	Md

Abbreviations: a discrimination classification, *b* difficulty classification, AL Joy, *BA* open mouth, *BF* closed mouth, *E* Easy, *F* female, *H* High, *M* male, *Md* Moderate, *ME* fear, *NE* = neutral, *NO* disgust, *RA* anger, *SU* surprise, *TR* sadness, *VE* Very Easy, *VH* Very high

Task operationalization

The task consists of two distinct stages. Stage 1 is mandatory, whereas Stage 2 is optional and is intended to increase task difficulty and, consequently, its discriminatory power.

Stage 1 consists of 28 stimuli, comprising four faces for each emotional category and four neutral faces. Half of the faces depict female actors and half depict male actors. These stimuli correspond to the original posed emotional expressions, representing the maximum (100%) intensity of each emotion. Scores for this stage are calculated separately for each target emotion and range from 0 to 4, with higher scores indicating greater ability to recognize facial expressions of emotion.

The initial reliability of Stage 1 of the TAREFE was assessed using the Omega coefficient. The observed values were 0.81 for neutral, 0.78 for surprise, 0.77 for joy, 0.74 for fear, 0.71 for sadness, 0.67 for anger, and 0.55 for disgust, indicating internal consistency ranging from acceptable to excellent (Baker & Kim, 2018; McDonald, 1999).

Stage 2 consists of 48 stimuli. These stimuli—four from each of the six basic emotions—were subjected to a morphing procedure (i.e., a transition between two images) with the corresponding neutral face to generate images with reduced-intensity emotional expressions. The *FantaMorph* 5 software (version 5.4.8; Abrosoft, 2016) was used for this purpose. For each face, two morphed images were created with levels defined according to the literature in the field (Claudino et al., 2025; de Almondes et al., 2020; Torro-Alves et al., 2013): one comprising 70% of the emotional expression and 30% of the neutral expression, and another comprising equal proportions of emotional and neutral expressions (50% and 50%). Stimuli in Stage 2 are presented and scored in the same manner as in Stage 1. Response time is also recorded.

TAREFE is administered via an online platform. Initially, an instruction screen is displayed, followed by two unscored practice examples. After this training phase, Stage 1 begins, during which the 28 stimuli are presented in random order by the software. Responses (correct or incorrect) and response times are automatically recorded. At the end of Stage 1, a screen indicates that the stage has been completed and allows the participant to either terminate the task or proceed to Stage 2; this option is determined by the researcher or clinician, depending on the assessment objectives. If the task is terminated, a screen informs the participant that the task has ended. If Stage 2 is initiated, the 48 stimuli are presented in random order, with responses and response times recorded automatically. Upon completion, a final task-closing screen is displayed.

Figure 1 presents examples of the TAREFE, showing faces displaying different intensities of emotional expression.

Discussion

TAREFE is a computerized instrument developed to assess facial emotion recognition (FER) using static facial expressions portrayed by actors from diverse racial backgrounds, representing basic emotions. The instrument consists of two stages that differ in the emotional intensity of the stimuli presented, a strategy designed to increase measurement sensitivity, particularly in typical populations and/or in individuals with less pronounced impairments in FER processes.

Its operationalization was designed to meet as many requirements and recommendations from the specialized literature as possible, with the aim of avoiding or minimizing biases and flaws in measurement estimation. First, the use of a computerized instrument aligns with contemporary trends in the incorporation of digital technologies into psychological assessment in Brazil, enabling—when appropriate technical and ethical criteria are met—both remote online administration and in-person application, thereby broadening the scope of assessment processes (Marasca et al., 2020). In addition, computerized instruments promote greater standardization of data collection procedures and reduce biases related to presentation order, learning, and habituation effects by allowing randomized stimulus presentation. Moreover, they enable faster processing of results, reduce the evaluator's workload, and increase accuracy by eliminating potential human errors associated with manual scoring.

The selection of the photographic stimulus set was conducted using a sample composed exclusively of university students, which did not compromise the analyses, as adequate dispersion of raw scores and sufficient variability across the latent trait continuum were observed, allowing the convergence of the estimation algorithms without evidence of parameter bias. Important criteria were taken into account, including balance in the representation of Brazil's ethnic and racial diversity among the selected actors., including balance in the representation of Brazil's ethnic and racial diversity among the selected actors. Despite our efforts, there was a predominance of white faces. Such variables have been shown to influence the FER process, either because people tend to recognize emotions more accurately in faces of the same ethnicity (Craig et al., 2017) or because culture modulates both the expression of emotions—through so-called display rules—and their perception, via differences in cognitive styles involved in interpretation (Engelmann & Pogosyan, 2013). The stimuli produced by Rodrigues (2015) are characterized by the use of Brazilian actors with diverse racial backgrounds.

Furthermore, the stimuli derived from facial databases followed a rigorous standardization protocol, with control of environmental conditions, suppression of distracting elements, and inclusion of a neutral facial expression, in accordance with previous recommendations (Fabrício et al., 2022). The stimuli are presented in color, which enhances the ecological validity of the task, and maintain gender balance among actors—a relevant consideration given evidence of variability in response patterns to faces that are gender-congruent or incongruent with the respondent (Hofmann et al., 2006). Additionally, the inclusion of variations in mouth position (open *vs.* closed) represents an important methodological refinement, as emotional expressions with an open mouth have greater potential to capture automatic attention than those with a closed mouth (Lageslag et al., 2018).

However, a limitation of this stimulus set is the restriction to the young adult age group. According to the literature (Borod et al., 2004; Ebner & Johnson, 2009; Ebner et al., 2013), FER can be influenced by the age of the face depicted in the stimulus. Specifically, accuracy tends to be lower for older adult faces than for young adult faces, as the latter provide more salient emotional cues. In addition, the age of the face may interact with the age of the respondent, further influencing recognition performance. Therefore, the impact of age-related variables on the response process should be examined in future psychometric studies of the instrument, particularly through Differential Item Functioning (DIF) analyses.

It is noteworthy that, according to the systematic review conducted by Fabrício et al. (2022), which examined the literature published up to 2020, the availability of Brazilian stimulus sets was limited to only four, all of which

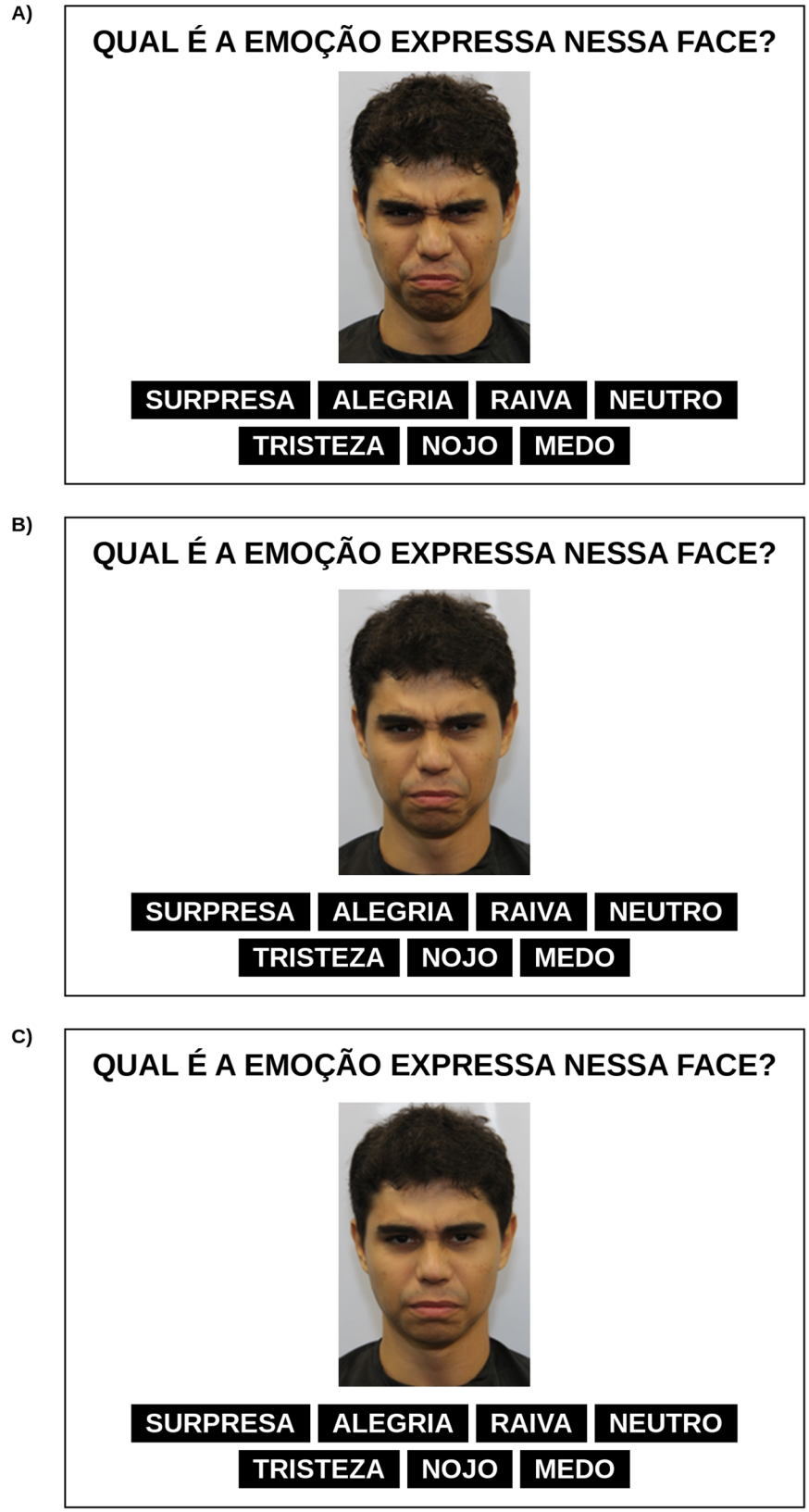


Fig. 1 Illustrative example of the TAREFE showing different intensities of a facial expression of disgust. Legend: **A**) 100% emotional intensity; **B**) 70% emotional intensity; **C**) 50% emotional intensity

were designed for child and adolescent populations (Donadon et al., 2019; Negrão et al., 2021; Novello et al., 2018; Romani-Sponchiado et al., 2015). More recently, in addition to the stimulus set developed by Rodrigues (2015), a new set was proposed by Morais et al. (2024), including actors from different age groups. Although this latter stimulus set offers greater age variability and other relevant technical attributes, it was not adopted in the present study due to its limited number of images, which could compromise both the final composition of the task and its psychometric viability in IRT-based analyses.

Among the studies analyzed by Fabrício et al. (2022), fewer than 10% employed more sophisticated techniques, such as IRT, for the inclusion of faces in stimulus banks. The most commonly used methods were accuracy rates, typically including stimuli with 70% or higher accuracy, and interrater reliability. For the selection of stimuli composing the TAREFE, a more refined statistical approach was adopted: the two-parameter IRT model (2PL). This choice is justified by the fact that, unlike accuracy-based criteria, IRT enables item selection based on parameters that are independent of the sample, allowing the simultaneous estimation of item difficulty and discriminatory power. Moreover, IRT provides information on each item's contribution to measurement precision across the latent trait continuum, as well as fit indices that assist in identifying and excluding poorly functioning or uninformative items. In this way, as argued by Ayala (2009), more precise and theoretically consistent psychometric instruments can be developed.

As can be observed, many stimuli were not selected for the final version of the TAREFE due to low or absent response variability, inadequate fit indices, and/or low difficulty levels. The large number of stimuli with low difficulty is likely associated with the high accuracy rates reported in the study by Rodrigues (2015), greater than 70% for sadness, joy, anger, neutral, surprise, and disgust, and close to 40% for fear. This distribution limited the inclusion of highly difficult stimuli, with 11% of the items classified as very easy, 32% easy, and 57% as having moderate difficulty. This limitation was partially mitigated by prioritizing items with greater discriminatory power. Overall, discriminatory power levels were considered adequate, with 25% of the stimuli showing very high discriminatory power, 29% high discriminatory power, and 46% moderate discriminatory power.

The stimuli were presented in a static format, although the literature reports ongoing debate regarding the optimal presentation modality. Some authors argue that dynamic stimuli provide richer information about emotional intensity and authenticity (Alves, 2013; Barrett et al., 2019). Nevertheless, empirical findings are mixed, with some studies reporting higher recognition rates for static stimuli and others finding no significant differences between formats (Fabrício et al., 2022). In a comparative study, Khosdelazad et al. (2020) examined the performance of healthy participants across three test formats—static images, dynamic morphing of photographs, and video stimuli—and concluded that each format captures a distinct aspect of the construct. In that study, dynamic presentation was identified as the most challenging format, with performance being influenced by variables such as age and educational level.

In this same vein, static stimuli were presented at 100% intensity. Given the overall difficulty profile of the items—specifically, the absence of highly difficult stimuli—a second, optional stage of the task is included, in which stimuli are presented at lower intensity levels (50% and 70%). This approach is intended to increase task difficulty and reduce the likelihood of ceiling effects in studies involving samples with subtle impairments in FER. According to Wells et al. (2016), in everyday life, emotional expressions are typically conveyed at intermediate intensity levels (low to moderate); therefore, the use of expressions with varying intensity enhances ecological validity and may increase the sensitivity of FER tasks to subtle individual differences. In an empirical study, these authors found that recognition accuracy was higher for more intense expressions, whereas response times were shorter for expressions of moderate intensity. This stimulus presentation format is consistent with similar methodological approaches in other contexts (Shimizu et al., 2025). It should be noted, however, that this stage of the TAREFE still requires psychometric studies to evaluate its adequacy and incremental validity.

Finally, multiple performance metrics were employed to provide a more comprehensive and sensitive assessment of FER, capturing its distinct processing components. For example, a review of the literature on facial emotion processing in individuals with depression identified alterations at different processing levels, including increased selective attention and vigilance toward sad faces and avoidance of happy faces, whereas evidence of

impaired recognition accuracy was less consistently observed (Bourke et al., 2010). These findings illustrate the complexity of FER and highlight the limitations of measures based exclusively on accuracy.

In conclusion, TAREFE appears to be a promising instrument. Its initial psychometric indicators, obtained through contemporary statistical techniques, proved to be robust, with items covering different levels of difficulty and demonstrating appropriate discriminatory power.

Notwithstanding the strength of these indicators, further studies are underway to expand evidence of the instrument's validity and reliability and to support its adaptation for clinical use, in accordance with Brazilian regulations for psychological tests (SATEPSI; CFP, 2022). Ongoing investigations include: (a) further empirical analyses of the internal structure of TAREFE; (b) evaluation of item difficulty, discriminatory power and incremental validity for Stage 2 stimuli; (c) Differential Item Functioning (DIF) analyses considering sex, age, and race; (d) development of normative data for the Brazilian population; (d) analysis of diagnostic accuracy (sensitivity and specificity) for neurocognitive disorders, autism spectrum disorder, and schizophrenia; and (e) assessment of temporal stability.

Thus, TAREFE is a promising tool applicable to both typical samples and clinical populations and can enrich psychological and neuropsychological assessment across different contexts, providing support for both the diagnostic process and the planning and monitoring of therapeutic interventions.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1186/s41155-026-00410-4.

Acknowledgements The authors thank Ana Luiza Fogaça da Mota, undergraduate research student, for her assistance with the morphing procedures.

Authors' contributions FLO and NTA made substantial contributions to the conception and design of the study, data analysis, drafting the manuscript, and critical revision for important intellectual content. DAF made substantial contributions to data analysis, drafting of the manuscript, and critical revision for important intellectual content. All authors approved the final version of the manuscript.

Data availability The datasets used and/or analyzed in this study are available from the corresponding author upon request.

Declarations

Ethics approval and consent to participate This study was conducted in accordance with the ethical principles of the Brazilian National Health Council Resolutions CNS nº 466/2012 and nº 510/2016. The research project was reviewed and approved by the Institutional Review Board of the Universidade Federal da Paraíba—UFPB (CAAE: 03594912.1.0000.5188) for the stimulus development phase, and by the Ethics Committee of Hospital das Clínicas da Faculdade de Medicina de Ribeirão Preto da USP for the analytical process assessment (CAEE: 60917822.6.0000.5440). All participants were provided with detailed information regarding the study's justification, objectives, procedures, risks, and benefits through an Informed Consent Form (Termo de Consentimento Livre e Esclarecido—TCLE). All participants provided written informed consent to participate in the study and to have their images used for research purposes. Participation was entirely voluntary, and participants were explicitly informed of their right to withdraw their consent at any stage of the research without facing any penalty or discomfort.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Abrosoft (2016). Fantamorph version 5.4.8 [Computer software]. Retrieved from http://www.fantamorph.com/.

Alves, N. T. (2013). Recognition of static and dynamic facial expressions: A study review. *Estudos De Psicologia, 18*(1), 125–130. https://doi.org/10.1590/s1413-294x2013000100020

Andrade, DFde, & Valle, RdaC. (1998). Introdução à Teoria da Resposta ao Item. *Estudos Em Avaliação Educacional, 18*, 13–32. https://doi.org/10.18222/eae01819982250

Andrade, D. F., Tavares, H. R., & da Cunha Valle, R. (2000). *Teoria da Resposta ao Item: Conceitos e aplicações* (p. 154). ABE.

Andrade JM, Laros JA, and Lima KS. 2021. Teoria de resposta ao item paramétrica e não paramétrica. In: Faiad C, Baptista MN, and Primi R, eds. Tutoriais em Análise de Dados Aplicados à Psicometria. Petrópolis - RJ: Editora Vozes.

Ayala, R. J. (2009). *The theory and practice of item response theory*. Guilford Publications.

de Ayala, R. J. (2013). The theory and practice of item response theory. Guilford Publications.

Baker, F. B., & Kim, S. H. (2017). *The basics of item response theory using R* (Vol. 10, pp. 1–174). Springer. https://doi.org/10.1007/978-3-319-54205-8

Baker, F. B., & Kim, S.-H. (2018). *The basics of item response theory using R*. Springer International Publishing.

Barrett, L. F., Adolphs, R., Marsella, S., Martinez, A. M., & Pollak, S. D. (2019). Emotional expressions reconsidered: Challenges to inferring emotion from human facial movements. *Psychological Science in the Public Interest: A Journal of the American Psychological Society, 20*(1), 1–68. https://doi.org/10.1177/1529100619832930

Bond, T., Yan, Z., & Heene, M. (2020). *Applying the Rasch model: Fundamental measurement in the human sciences*. Routledge.

Boone, W. J., Staver, J. R., & Yale, M. S. (2013). *Rasch analysis in the human sciences* (Vol. 10, pp. 978–994). Springer. https://doi.org/10.1007/978-94-007-6857-4

Boone WJ. (2016). Rasch Analysis for Instrument Development: Why, When, and How? CBE Life Sci Educ 15. https://doi.org/10.1187/cbe.16-04-0148

Borod, J. C., Yecker, S. A., Brickman, A. M., Moreno, C. R., Sliwinski, M., Foldi, N. S., Alpert, M., & Welkowitz, J. (2004). Changes in posed facial expression of emotion across the adult life span. *Experimental Aging Research, 30*(4), 305–331. https://doi.org/10.1080/03610730490484399

Bourke, C., Douglas, K., & Porter, R. (2010). Processing of facial emotion expression in major depression: A review. *The Australian and New Zealand Journal of Psychiatry, 44*(8), 681–696. https://doi.org/10.3109/00048674.2010.496359

Chen, W.-H., & Thissen, D. (1997). Local dependence indexes for item pairs using item response theory. *Journal of Educational and Behavioral Statistics: A Quarterly Publication Sponsored by the American Educational Research Association and the American Statistical Association, 22*(3), 265–289. https://doi.org/10.3102/10769986022003265

Claudino, RGe., da Costa, I. F., de Nascimento, INdeA., & Torro, N. (2020). Use of sign language does not favor recognition of static and dynamic emotional faces in deaf people. *Psychology & Neuroscience, 13*(4), 531–538. https://doi.org/10.1037/pne0000238

Claudino, RGe., Garcia, R. B., & Torro-Alves, N. (2025). Hypervigilance to dynamic and static facial expressions in social anxiety: An eye-tracking study. *Psychology & Neuroscience, 18*(4), 338–348. https://doi.org/10.1037/pne0000369

Conley, M. I., Dellarco, D. V., Rubien-Thomas, E., Cohen, A. O., Cervera, A., Tottenham, N., & Casey, B. J. (2018). The racially diverse affective expression (RADIATE) face stimulus set. *Psychiatry Research, 270*, 1059–1067. https://doi.org/10.1016/j.psychres.2018.04.066

Conselho Federal de Psicologia. (2022). *SATEPSI*. SATEPSI. https://satepsi.cfp.org.br/

Craig, B. M., Zhang, J., & Lipp, O. V. (2017). Facial race and sex cues have a comparable influence on emotion recognition in Chinese and Australian participants. *Attention, Perception & Psychophysics, 79*(7), 2212–2223. https://doi.org/10.3758/s13414-017-1364-z

Darwin, C. (1872/1998). *The expression of the emotions in man and animals* (Paul Ekman, Ed.; 3rd ed.). Oxford University Press

de Almondes, K. M., Júnior, F. W. N. H., Leonardo, M. E. M., & Alves, N. T. (2020). Facial emotion recognition and executive functions in insomnia disorder: An exploratory study. *Frontiers in Psychology, 11*, Article 502. https://doi.org/10.3389/fpsyg.2020.00502

Donadon, M. F., Martin-Santos, R., & Osório, F. L. (2019). Baby faces: Development and psychometric study of a stimuli set based on babies' emotions. *Journal of Neuroscience Methods, 311*, 178–185. https://doi.org/10.1016/j.jneumeth.2018.10.021

Ebner, N. C., & Johnson, M. K. (2009). Young and older emotional faces: Are there age group differences in expression identification and memory? *Emotion, 9*(3), 329–339. https://doi.org/10.1037/a0015179

Ebner, N. C., Riediger, M., & Lindenberger, U. (2010). FACES–a database of facial expressions in young, middle-aged, and older women and men: Development and validation. *Behavior Research Methods, 42*(1), 351–362. https://doi.org/10.3758/BRM.42.1.351

Ebner, N. C., Johnson, M. R., Rieckmann, A., Durbin, K. A., Johnson, M. K., & Fischer, H. (2013). Processing own-age vs. other-age faces: Neuro-behavioral correlates and effects of emotion. *NeuroImage, 78*, 363–371. https://doi.org/10.1016/j.neuroimage.2013.04.029

Ekman, P. (1972). *Emotion in the human face: Guide-lines for research and an integration of findings*. Pergamon.

Ekman, P. (1993). Facial expression and emotion. *American Psychologist, 48*(4), 384–392. https://doi.org/10.1037/0003-066x.48.4.384

Ekman, P. (2009). Darwin's contributions to our understanding of emotional expressions. *Philosophical Transactions of the Royal Society of London. Series b, Biological Sciences, 364*(1535), 3449–3451. https://doi.org/10.1098/rstb.2009.0189

Ekman, P., & Friesen, W. V. (1969). The repertoire of nonverbal behavior: Categories, origins, usage, and coding. *Semiotica, 1*(1), 49–98. https://doi.org/10.1515/semi.1969.1.1.49

Ekman, P., & Friesen, W. V. (1971). Constants across cultures in the face and emotion. *Journal of Personality and Social Psychology, 17*(2), 124–129. https://doi.org/10.1037/h0030377

Ekman, P., & Friesen, W. V. (1976). *Pictures of facial affect*. Consulting Psychologists Press.

Engelmann, J. B., & Pogosyan, M. (2013). Emotion perception across cultures: The role of cognitive mechanisms. *Frontiers in Psychology, 4*, Article 118. https://doi.org/10.3389/fpsyg.2013.00118

Fabrício, DdeM., Ferreira, B. L. C., Maximiano-Barreto, M. A., Muniz, M., & Chagas, M. H. N. (2022). Construction of face databases for tasks to recognize facial expressions of basic emotions: A systematic review. *Dementia & Neuropsychologia, 16*(4), 388–410. https://doi.org/10.1590/1980-5764-dn-2022-0039

Gao, Z., Zhao, W., Liu, S., Liu, Z., Yang, C., & Xu, Y. (2021). Facial emotion recognition in schizophrenia. *Frontiers in Psychiatry, 12*, Article 633717. https://doi.org/10.3389/fpsyt.2021.633717

Happy, S. L., Patnaik, P., Routray, A., & Guha, R. (2017). The Indian spontaneous expression database for emotion recognition. *IEEE Transactions on Affective Computing, 8*(1), 131–142. https://doi.org/10.1109/taffc.2015.2498174

He, W., Chai, H., Chen, W., Zhang, J., Xu, Y., Zhu, J., & Wang, W. (2012). Facial emotion triggered cerebral potentials in treatment-resistant depression and borderline personality disorder patients of both genders. *Progress in Neuro-Psychopharmacology and Biological Psychiatry, 37*(1), 121–127. https://doi.org/10.1016/j.pnpbp.2011.12.003

Hofmann, S. G., Suvak, M., & Litz, B. T. (2006). Sex differences in face recognition and influence of facial affect. *Personality and Individual Differences, 40*(8), 1683–1690. https://doi.org/10.1016/j.paid.2005.12.014

Holmes Finch, W., & French, B. F. (2015). *Latent variable modeling with R*. Routledge.

Huang, Y.-L., Chen, T.-T., Wang, W.-S., Kuo, C. Y., Yang, Y. K., & Tseng, H.-H. (2025). Low versus high levels of social cognition impairment and their associations with specific schizophrenia symptom domains. *The Journal of Neuropsychiatry and Clinical Neurosciences, 37*(3), 261–269. https://doi.org/10.1176/appi.neuropsych.20240020

Kaulard, K., Cunningham, D. W., Bülthoff, H. H., & Wallraven, C. (2012). The MPI facial expression database--A validated database of emotional and conversational facial expressions. *PLoS ONE, 7*(3), Article e32321. https://doi.org/10.1371/journal.pone.0032321

Khosdelazad, S., Jorna, L. S., McDonald, S., Rakers, S. E., Huitema, R. B., Buunk, A. M., & Spikman, J. M. (2020). Comparing static and dynamic emotion recognition tests: Performance of healthy participants. *PLoS ONE, 15*(10), Article e0241297. https://doi.org/10.1371/journal.pone.0241297

Kim, S.-M., Kwon, Y.-J., Jung, S.-Y., Kim, M.-J., Cho, Y. S., Kim, H. T., Nam, K.-C., Kim, H., Choi, K.-H., & Choi, J.-S. (2017). Development of the Korean facial emotion stimuli: Korea University Facial Expression Collection 2nd edition. *Frontiers in Psychology, 8*, Article 769. https://doi.org/10.3389/fpsyg.2017.00769

Kohler, C. G., Turner, T. H., Gur, R. E., & Gur, R. C. (2004). Recognition of facial emotions in neuropsychiatric disorders. *CNS Spectrums, 9*(4), 267–274. https://doi.org/10.1017/s1092852900009202

Krause, F. C., Linardatos, E., Fresco, D. M., & Moore, M. T. (2021). Facial emotion recognition in major depressive disorder: A meta-analytic review. *Journal of Affective Disorders, 293*, 320–328. https://doi.org/10.1016/j.jad.2021.06.053

Langeslag, S. J. E., Gootjes, L., & van Strien, J. W. (2018). The effect of mouth opening in emotional faces on subjective experience and the early posterior negativity amplitude. *Brain and Cognition, 127*, 51–59. https://doi.org/10.1016/j.bandc.2018.10.003

Lee, S. B., Koo, S. J., Song, Y. Y., Lee, M. K., Jeong, Y.-J., Kwon, C., Park, K. R., Park, J. Y., Kang, J. I., Lee, E., & An, S. K. (2014). Theory of mind as a mediator of reasoning and facial emotion recognition: Findings from 200 healthy people. *Psychiatry Investigation, 11*(2), 105–111. https://doi.org/10.4306/pi.2014.11.2.105

Lin, Y.-H., Hsin, Y.-L., Li, R.-H., Liu, C.-K., Wang, R.-Y., & Wang, W.-H. (2022). The effect of facial expression intensity on emotion recognition and psychosocial performance in patients with frontal or temporal lobe epilepsy. *Epilepsy & Behavior: E&b, 126*(108462), Article 108462. https://doi.org/10.1016/j.yebeh.2021.108462

Linacre, J. M. (2002). What do infit and outfit, mean-square and standardized mean. Rasch measurement transactions, 16(2), 878. URL: www.rasch.org/rmt/rmt162f.htm

LoBue, V., & Thrasher, C. (2014). The Child Affective Facial Expression (CAFE) set: Validity and reliability from untrained adults. *Frontiers in Psychology, 5*, 1532. https://doi.org/10.3389/fpsyg.2014.01532

Ma, J., Yang, B., Luo, R., & Ding, X. (2020). Development of a facial-expression database of Chinese Han, Hui and Tibetan people. *International Journal of Psychology. Journal International De Psychologie, 55*(3), 456–464. https://doi.org/10.1002/ijop.12602

MacPherson, H. A., Kudinova, A. Y., Jenkins, G. A., Kim, K. L., Radoeva, P. D., Gilbert, A. C., Barthelemy, C., DeYoung, L., Yen, S., Hower, H., Hunt, J., Keller, M. B., & Dickstein, D. P. (2021). Facial emotion recognition and mood symptom course in young adults with childhood-onset bipolar disorder. *European Archives of Psychiatry and Clinical Neuroscience, 271*(7), 1393–1404. https://doi.org/10.1007/s00406-021-01252-0

Marais, I., & Andrich, D. (2008). Effects of varying magnitude and patterns of response dependence in the unidimensional Rasch model. *Journal of Applied Measurement, 9*(2), 105–124.

Marasca, A. R., Yates, D. B., Schneider, AMdeA., Feijó, L. P., & Bandeira, D. R. (2020). Avaliação psicológica online: Considerações a partir da pandemia do novo coronavírus (COVID-19) para a prática e o ensino no contexto a distância. *Estudos De Psicologia (Campinas)*. https://doi.org/10.1590/1982-0275202037e200085

McDonald, R. P. (1999). *Test theory: A unified treatment*. Psychology Press.

Meehan, K. B., De Panfilis, C., Cain, N. M., Antonucci, C., Soliani, A., Clarkin, J. F., & Sambataro, F. (2017). Facial emotion recognition and borderline personality pathology. *Psychiatry Research, 255*, 347–354. https://doi.org/10.1016/j.psychres.2017.05.042

Miotto, E. C., Campanholo, K. R., Serrao, V. T., & Trevisan, B. T. (2023). *Manual de Avaliação Psicológica - Volume 1 & 2*. MEMNON.

Montagne, B., Kessels, R. P. C., De Haan, E. H. F., & Perrett, D. I. (2007). The emotion recognition task: A paradigm to measure the perception of facial emotional expressions at different intensities. *Perceptual and Motor Skills, 104*(2), 589–598. https://doi.org/10.2466/pms.104.2.589-598

Morais, D., Maximiano-Barreto, M. A., Luchesi, B. M., Muniz, M., & Chagas, M. H. N. (2024). Development and validation of a face database for the recognition of facial expressions of basic emotions in the Brazilian population. *Psicologia, 37*(1), Article 47. https://doi.org/10.1186/s41155-024-00325-y

Muñoz Ladrón de Guevara, C., Reyes Del Paso, G. A., Fernández-Serrano, M. J., & Duschek, S. (2021). Facial emotion recognition and executive functions in fibromyalgia. *Pain Medicine, 22*(7), 1619–1629. https://doi.org/10.1093/pm/pnab024

Negrão, J. G., Osorio, A. A. C., Siciliano, R. F., Lederman, V. R. G., Kozasa, E. H., D'Antino, M. E. F., Tamborim, A., Santos, V., de Leucas, D. L. B., Camargo, P. S., Mograbi, D. C., Mecca, T. P., & Schwartzman, J. S. (2021). The child emotion facial expression set: A database for emotion recognition in children. *Frontiers in Psychology, 12*, Article 666245. https://doi.org/10.3389/fpsyg.2021.666245

Novello, B., Renner, A., Maurer, G., Musse, S., & Arteche, A. (2018). Development of the youth emotion picture set. *Perception*. https://doi.org/10.1177/0301006618797226

Nowicki, S., Jr., & Carton, J. (1993). The measurement of emotional intensity from facial expressions. *Journal of Social Psychology, 133*(5), 749–750. https://doi.org/10.1080/00224545.1993.9713934

Oliver, L. D., Haltigan, J. D., Gold, J. M., Foussias, G., DeRosse, P., Buchanan, R. W., Malhotra, A. K., Voineskos, A. N., SPINS Group. (2019). Lower- and higher-level social cognitive factors across individuals with schizophrenia spectrum disorders and healthy controls: Relationship with neurocognition and functional outcome. *Schizophrenia Bulletin, 45*(3), 629–638. https://doi.org/10.1093/schbul/sby114

Orlando, M., & Thissen, D. (2003). Further investigation of the performance of S-X^2: An item fit index for use with dichotomous item response theory models. *Applied Psychological Measurement, 27*(4), 289–298. https://doi.org/10.1177/0146621603027004004

Quinting, J., Jonas, K., Kuhn, C., & Stenneken, P. (2022). Emotion recognition, empathy, or ToM? The influence of social cognition on communication in traumatic brain injury: A systematic review. *Zeitschrift Für Neuropsychologie, 33*(2), 59–69. https://doi.org/10.1024/1016-264x/a000355

Revilla, R., Nelson, C. M., Heilman, M., Prothro, K., Cheever, A., Hersey, J., Dede, B., & Hudac, C. M. (2025). Facial emotion recognition trainings for children and adolescents: A transdiagnostic systematic review and meta-analysis. *Clinical Psychology & Psychotherapy, 32*(4), Article e70116. https://doi.org/10.1002/cpp.70116

Rodrigues, M. R. (2015). Composição de um banco de expressões faciais brasileiro: um estudo de validação e comparação transcultural. [Dissertação de Mestrado - Universidade Federal da Paraíba]. URL: https://repositorio.ufpb.br/jspui/handle/tede/7768

Romani-Sponchiado, A., Sanvicente-Vieira, B., Mottin, C., Hertzog-Fonini, D., & Arteche, A. (2015). Child emotions picture set (CEPS): Development of a database of children's emotional expressions. *Psychology & Neuroscience, 8*(4), 467–478. https://doi.org/10.1037/h0101430

Sartes, L. M. A., & Souza-Formigoni, M. L. O. D. (2013). Avanços na psicometria: Da teoria clássica dos testes à teoria de resposta ao item. *Psicologia: Reflexão e Crítica, 26*, 241–250. https://doi.org/10.1590/S0102-79722013000200004

Schlegel, K., Grandjean, D., & Scherer, K. R. (2014). Introducing the Geneva emotion recognition test: An example of Rasch-based test development. *Psychological Assessment, 26*(2), 666–672. https://doi.org/10.1037/a0035246

Shimizu, Y., Ogawa, K., Kimura, M., Fujiwara, K., & Watanabe, N. (2024). The influence of emotional facial expression intensity on decoding accuracy: High intensity does not yield high accuracy. *The Japanese Psychological Research, 66*(4), 521–540. https://doi.org/10.1111/jpr.12529

Shimizu, Y., Ogawa, K., Kimura, M., Fujiwara, K., & Watanabe, N. (2025). Introducing the Facial Expressions of emotion decoding Test with various Intensity (FEET -I): Using the AIST Facial Expression Database 2017. *The Japanese Psychological Research*, *jpr.70009*. https://doi.org/10.1111/jpr.70009

Steephen, J. E., Kummetha, S., Obbineni, S. C., & Bapi, R. S. (2021). Mood-congruent biases in facial emotion perception and their gender dependence. *International Journal of Psychology: Journal International De Psychologie, 56*(3), 378–386. https://doi.org/10.1002/ijop.12720

Tholl, S., Sojer, C. A., Schmidt, S. N. L., & Mier, D. (2024). How to elicit a negative bias? Manipulating contrast and saturation with the facial emotion salience task. *Frontiers in Psychology, 15*, Article 1284595. https://doi.org/10.3389/fpsyg.2024.1284595

Thonse, U., Behere, R. V., Praharaj, S. K., & Sharma, P. S. V. N. (2018). Facial emotion recognition, socio-occupational functioning and expressed emotions in schizophrenia versus bipolar disorder. *Psychiatry Research, 264*, 354–360. https://doi.org/10.1016/j.psychres.2018.03.027

Torro-Alves, N., Bezerra, I., Claudino, R. G. E., & Pereira, T. (2013). Influences of sex, type and intensity of emotion in the ecognition of static and dynamic facial expressions. Avances En Psicología Latinoamericana, 31(1), 192–199. http://www.scielo.org.co/scielo.php?script=sci_arttext&pid=S1794-47242013000100016&lng=en&tlng=en.

Tottenham, N., Tanaka, J. W., Leon, A. C., McCarry, T., Nurse, M., Hare, T. A., Marcus, D. J., Westerlund, A., Casey, B. J., & Nelson, C. (2009). The NimStim set of facial expressions: Judgments from untrained research participants. *Psychiatry Research, 168*(3), 242–249. https://doi.org/10.1016/j.psychres.2008.05.006

Vikhanova, A., Mareschal, I., & Tibber, M. (2022). Emotion recognition bias depends on stimulus morphing strategy. *Attention, Perception & Psychophysics, 84*(6), 2051–2059. https://doi.org/10.3758/s13414-022-02532-0

Wells, L. J., Gillespie, S. M., & Rotshtein, P. (2016). Identification of emotional facial expressions: Effects of expression, intensity, and sex on eye gaze. *PLoS ONE, 11*(12), Article e0168307. https://doi.org/10.1371/journal.pone.0168307

Williams, T. F., Vehabovic, N., & Simms, L. J. (2023). Developing and validating a facial emotion recognition task with graded intensity. *Assessment, 30*(3), 761–781. https://doi.org/10.1177/10731911211068084

Wu, M., Tam, H. P., & Jen, T. (2016). *Educational measurement for applied researchers: Theory into practice*. Springer.

Yen, W. M. (1984). Effects of local item dependence on the fit and equating performance of the three-parameter logistic model. *Applied Psychological Measurement, 8*(2), 125–145. https://doi.org/10.1177/014662168400800201

Yen, W. M. (1993). Scaling performance assessments: Strategies for managing local item dependence. *Journal of Educational Measurement, 30*(3), 187–213. https://doi.org/10.1111/j.1745-3984.1993.tb00423.x

Young, A. W., Rowland, D., Calder, A. J., Etcoff, N. L., Seth, A., & Perrett, D. I. (1997). Facial expression megamix: Tests of dimensional and category accounts of emotion recognition. *Cognition, 63*(3), 271–313. https://doi.org/10.1016/s0010-0277(97)00003-6

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Flávia de Lima Osório[1,2] · Diego Alves Ferreira[1] · Nelson Torro-Alves[3]

✉ Flávia de Lima Osório
flaliosorio@gmail.com

Diego Alves Ferreira
diegoaferreira@usp.br

Nelson Torro-Alves
nelsontorro@gmail.com

[1] Departamento de Neurociências e Ciências do Comportamento, Universidade de São Paulo, São Paulo, Brazil

[2] (INCT-MT), National Institute of Science and Technology for Translational Medicine, Brasília, Brazil

[3] Departamento de Psicologia, Federal University of Paraíba, João Pessoa, Brazil